

# llmXive Automated Review of MobileForge: Annotation-Free Adaptation for Mobile GUI Agents with Hierarchical Feedback-Guided Policy Optimization

*llmXive automated review panel*

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**Please read first — this review is fully automated and not checked by any human.** It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

## How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

**This paper’s panel (9 lenses):** Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

## Claim Accuracy

*Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_claim\_accuracy.md

The review focuses on the accuracy of factual claims and their alignment with the provided data and citations.

**1. Mathematical Precision of Claims** In Section 4.2 (Cross-Domain Generalization), the text states: “The adapted \llmname{ForgeOwl-8B} attains **41.0** \% success... achieving a **+9.0** \% relative gain over its 8B base.”

- **Verification:** Base score = 37.6% (Table 2). Adapted score = 41.0%.
- **Calculation:** Absolute gain = 3.4 percentage points. Relative gain =  $(41.0 - 37.6) / 37.6 = 9.04\%$ .
- **Assessment:** The math is correct. However, in scientific writing, “relative gain” is often confused with “percentage point gain.” The claim is accurate but risks misinterpretation. The text should explicitly clarify “9.0% relative improvement” to prevent readers from assuming a 9.0 percentage point increase.

**2. Consistency of Performance Claims** The Abstract and Section 4.1 claim the adapted Qwen3-VL-8B reaches **67.2% Pass@3**, close to the base GUI-Owl-1.5-8B (**69.0%**).

- **Verification:** Table 1 confirms Qwen3-VL-8B (900 tasks) Pass@3 is 67.2% and GUI-Owl-1.5-8B (0 tasks) is 69.0%.
- **Assessment:** The claim is accurate. The comparison is valid.

**3. Ablation Study Claims** In Section 4.3 (Task filtering), the text claims: “Retaining all-fail and mixed tasks ( $SR \in [0.0, 0.9]$ ) yields the best combined... performance.”

- **Verification:** Table 4 shows the  $[0.0, 0.9]$  range yields 48.3% (AndroidWorld) and 15/117 (MobileWorld), while the  $[0.0, 1.0]$  range yields 48.3% and 10/117.
- **Assessment:** The claim that removing all-fail tasks harms MobileWorld success is supported (15 vs 10). However, the claim that  $[0.0, 0.9]$  is the *best* combined performance relies on the assumption that no other intermediate range (e.g.,  $[0.1, 0.8]$ ) performs better. Since the table caption notes “representative rows shown,” the claim of “best” is slightly overstated without the full ablation curve.

**4. Citation Accuracy** The paper cites works like \citep{zhang2025tongui} to support the claim that manual annotation is costly. These citations appear consistent with the general consensus in the cited literature regarding the cost of human-written tasks in GUI agent research. No obvious misattribution is found.

**Conclusion** The factual claims are largely supported by the data. The primary issues are

minor potential ambiguities in phrasing (“relative gain”) and the strength of the “best” claim in the ablation study given the limited data shown.

#### Action items.

- **[writing]** Clarify the ‘+9.0% relative gain’ claim in Section 4.2. While mathematically correct (3.4/37.6), the phrasing risks confusion with absolute percentage points. Explicitly state ‘9.0% relative improvement’ to distinguish from the 3.4pp absolute gain.
- **[science]** In Section 4.3, the claim that the [0.0, 0.9] filter yields the ‘best’ performance relies on ‘representative rows’ in Table 4. Verify if the full ablation curve was checked or if other ranges (e.g., [0.1, 0.8]) might outperform, as the current data only compares two specific points.
- **[writing]** Ensure consistent distinction between ‘base’ and ‘adapted’ model scores in the Abstract and Section 4.1. The 69.0% figure is the base GUI-Owl score; ensure the text does not inadvertently imply this is the adapted score when comparing to the 67.2% adapted Qwen3 score.

## Figure Critic

*Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_figure\_critic.md

### Figure 1

The figure effectively illustrates the concept of corrective hints with clear visual steps, but the caption contains a grammatical fragment and the diagram labels include a spelling error.

### Figure 2

The figure effectively contrasts the failure of the base model with the success of the adapted model, but the visual evidence in panel (b) omits the deletion of the ‘Unexpected Expenses’ item mentioned in the caption.

### Figure 3

The heatmap effectively visualizes performance gains but suffers from readability issues due to small, rotated x-axis labels. Additionally, the y-axis labels and caption create ambiguity regarding the specific baseline used for the ‘failure-rate reduction’ calculation.

## Figure 4

The figure provides a clear visual workflow of the HiFPo method, but the caption is grammatically broken and lacks a clear subject, while the figure title contains a spelling error.

## Figure 5

The figure visually presents the MobileGym and HiFPO components but the caption is grammatically broken ('Overview of .'), and the diagram lacks explicit integration details showing how the two parts form the complete MobileForge system.

## Figure 6

The figure provides a clear visual overview of the MobileForge framework and its components, but the caption is severely degraded by missing subject and object names, making it grammatically incomplete and difficult to read.

## Figure 7

The figure provides a clear visual overview of the MobileGym framework, but the caption is grammatically broken with missing subject names, and the diagram itself contains a minor spelling error ('Depth-fist').

## Figure 8

The figure illustrates a failure trajectory in the Broccoli app, but the caption incorrectly identifies the task as 'ExpenseDeleteMultiple2' in the Pro Expense app, creating a fundamental mismatch between the visual evidence and the textual description.

## Figure 9

Figure 9 is labeled as a failure case but visually depicts a successful task completion with no clear indication of where the agent failed. The extensive red annotations and 'Failed' decision are not explained in the caption, making it unclear what specific failure is being demonstrated.

## Figure 10

The figure effectively visualizes a failure trajectory with clear step-by-step screenshots and annotations, but the caption is too brief to stand alone as a scientific record, failing to describe the task or the specific error shown.

## Figure 11

The figure displays a training reward curve but suffers from poor axis labeling, specifically a missing y-axis label and a poorly positioned x-axis label.

**Action items.**

- **[writing]** Figure 1: The caption contains a sentence fragment (‘summarizes the failure...’) that lacks a subject, likely referring to the ‘MobileGym-Critic’ module shown in the diagram; the text should be edited to explicitly name the subject.
- **[writing]** Figure 1: The labels ‘attampt 1’ and ‘attampt 2’ contain a spelling error (‘attampt’ instead of ‘attempt’).
- **[writing]** Figure 2: The caption states the task is to delete ‘Streaming Services, Unexpected Expenses, and Pet Supplies’, but the bottom panel (b) only shows the successful deletion of ‘Streaming Services’ and ‘Pet Supplies’. The ‘Unexpected Expenses’ item is missing from the visual workflow.
- **[science]** Figure 3: The y-axis labels (‘ForgeQwen3-8B vs. Qwen3-VL-8B’ and ‘ForgeOwl-8B vs. GUI-Owl-1.5-8B’) imply a comparison between two different base models, but the caption states the metric is ‘relative to the corresponding base agent’. This creates ambiguity: it is unclear if the ‘reduction’ is calculated against the specific base model listed in the label or a unified baseline, and the heatmap does not explicitly show the base model’s performance for context.
- **[writing]** Figure 3: The x-axis labels (e.g., ‘Req’, ‘Data entry’, ‘Compose UI’) are rotated 90 degrees and rendered in a font size that is difficult to read, reducing the figure’s accessibility.
- **[writing]** Figure 4: The caption text is grammatically incomplete and confusing, starting with ‘converts hierarchical feedback...’ and ‘performs hint-guided...’ without specifying the subject (likely ‘HiFPo’ or the method name), making it unclear what entity is performing these actions.
- **[writing]** Figure 4: The caption contains a typo in the title ‘Trajectory-Level Muti-Attempt Rollout’ (should be ‘Multi-Attempt’).
- **[fatal]** Figure 5: The caption reads ‘Overview of .’ with a missing subject, failing to identify the system (MobileForge) or components being overviewed.
- **[science]** Figure 5: The diagram shows ‘MobileGym’ and ‘HiFPO’ as separate blocks but lacks explicit arrows or text indicating how they combine to form ‘MobileForge’ as claimed in the title.
- **[writing]** Figure 6: The caption contains multiple grammatical gaps where the system name ‘MobileForge’ and component names ‘MobileGym’ and ‘HiFPO’ are missing (e.g., ‘Overview of .’, ‘combines for’, ‘with for’). These should be filled to match the labels in the figure.
- **[writing]** Figure 6: The caption text ‘Existing annotation-free GUI learning lacks a unified adaptation substrate’ is a sentence fragment that does not clearly connect to the visual elements without the missing subject ‘MobileForge’.
- **[writing]** Figure 7: The caption contains multiple grammatical errors and missing terms (e.g., ‘as the annotation-free adaptation substrate’, ‘grounds adaptation’, ‘through ,’, ‘with to produce’), likely due to missing bolded component names like ‘MobileGym’ or ‘MobileGym-Critic’.

- **[writing]** Figure 7: The diagram contains a typo in the ‘Target-App Exploration’ section, where ‘Depth-fist Traversal’ is written instead of ‘Depth-first Traversal’.
- **[science]** Figure 8: The caption states the task is ‘ExpenseDeleteMultiple2’ (deleting three expenses), but the figure shows the ‘Broccoli’ app and a task to delete three recipes. The visual content contradicts the caption’s description of the task and environment.
- **[writing]** Figure 8: The figure title ‘Trajectory of Qwen3-VL 8B’s Failure in AndroidWorld’ and the instruction text at the top are not defined in the caption, creating a disconnect between the visual narrative and the provided description.
- **[science]** Figure 9: The caption states ‘Qwen3-VL-8B base: failed’, but the image shows a successful completion of the task (graduates list found, email open ready to send) with no visible failure state or error message.
- **[writing]** Figure 9: The image contains extensive red annotations (step numbers, error bubbles, red X marks) and a ‘<Decision> Failed’ footer that are not defined or explained in the caption, making the failure mode unclear.
- **[science]** Figure 9: The trajectory shows the agent successfully navigating to the email inbox and finding the graduates list (steps 11-12), yet the figure is labeled as a failure without explaining where the actual failure occurred.
- **[science]** Figure 10: The figure displays a 14-step trajectory of a failed task, but the caption ‘GUI-Owl-1.5-8B base: failed’ is insufficient. It fails to describe the specific task (recalculating invoice payment) or the nature of the failure (calculation error), making the figure’s scientific contribution unclear without external context.
- **[writing]** Figure 10: The figure lacks a formal caption describing the visual content. The provided text ‘GUI-Owl-1.5-8B base: failed’ acts as a label rather than a descriptive caption, and the figure title ‘Trajectory of GUI-Owl-1.5 8B’s Failure in MobileWorld’ is not integrated into the standard caption format.
- **[writing]** Figure 11: The x-axis label ‘Step’ is positioned in the bottom-right corner without an axis line or tick marks, making it ambiguous whether it refers to the x-axis or is a stray label.
- **[writing]** Figure 11: The y-axis lacks a descriptive label (e.g., ‘Reward’ or ‘Score’); the title ‘reward/overall’ is present but axis labels are standard for clarity.

## Jargon Police

*Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_jargon\_police.md

The manuscript relies heavily on domain-specific acronyms and custom macros that obscure

meaning for a broader audience. The most critical issue is the use of ‘GRPO’ in Section 3.3 without defining it as Group Relative Policy Optimization. While experts in RLHF may recognize this, the paper claims to present a general adaptation system, and the method’s core relies on this specific algorithm.

Additionally, the frequent use of the custom macro `\ourmethod` (rendered as “MobileForge” in the final PDF but opaque in the source logic) and `\mobilegym` creates a barrier. In the Abstract and Introduction, the text should explicitly state the full name of the system and its components rather than relying on the reader to infer the expansion of these shorthand terms.

The term “substrate” is used repeatedly (e.g., “unified mobile substrate”) to describe the interaction environment. This is a metaphorical use of a biological/chemical term that may confuse readers outside of specific systems engineering contexts. Replacing this with “framework,” “platform,” or “environment” would be more precise and accessible.

Finally, technical training details in the Appendix mention “FSDP” and “KL regularization” without context. While “KL” is standard, “FSDP” should be spelled out at first use. The paper assumes a high level of prior knowledge regarding RLHF training mechanics (e.g., “clipped GRPO objective,” “advantage estimator”) which should be briefly contextualized for readers who are not RL specialists.

#### Action items.

- **[writing]** Define the acronym ‘GRPO’ (Group Relative Policy Optimization) at its first occurrence in Section 3.3. Currently, it appears as ‘GRPO’ without expansion, assuming reader familiarity with specific RLHF variants.
- **[writing]** Replace the custom macro ‘`\ourmethod`’ with the full system name ‘MobileForge’ in the Abstract and Introduction. While defined in LaTeX, the rendered text should not rely on the reader knowing the macro expansion to understand the subject.
- **[writing]** Define the term ‘substrate’ in the context of ‘unified mobile substrate’ (Section 1). The word is used metaphorically here; a brief clarification (e.g., ‘framework’ or ‘environment’) would aid non-specialist readers.
- **[writing]** Expand the acronym ‘FSDP’ (Fully Sharded Data Parallel) in the training details (Section 5/Appendix) if it appears for the first time there, or ensure it is defined earlier in the methodology.
- **[writing]** Clarify the term ‘curriculum’ in ‘Curriculum mining’ (Section 1). While standard in RL, explicitly stating it refers to ‘task difficulty scheduling’ or ‘progressive task selection’ would improve accessibility.

## Logical Consistency

*Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_logical\_consistency.md



The logical flow of the proposed MobileForge framework is generally sound, with a clear chain of reasoning from the problem setup (annotation-free adaptation) to the solution (MobileGym + HiFPO). The causal claims regarding the benefits of hierarchical feedback and hint-contextualized GRPO are well-supported by the ablation studies in Tables 3 and 4. Specifically, the removal of hint context leading to a 25pp drop in success (Table 3) strongly validates the mechanism of multi-attempt learning.

However, there are minor logical gaps in the interpretation of the ablation results that require clarification to ensure the conclusions strictly follow the premises:

1. **Task Filtering Logic (Section 3.3, Table 4):** The paper argues that retaining tasks with success rates (SR) in  $[0.0, 0.9]$  yields better out-of-domain (MobileWorld) performance than including all tasks  $[0.0, 1.0]$ . The logic implies that removing “mastered” tasks (SR=1) improves generalization. However, the table shows the “Medium + hard” filter has 1910 samples while “All tasks” has 2137. The conclusion that the *filtering strategy* (excluding easy tasks) is the cause of the MobileWorld gain (15/117 vs 10/117) is plausible but not strictly proven without controlling for the reduction in total training data. The authors should explicitly state whether the gain is attributed to the *quality* of the remaining data (focusing on hard cases) or if the “All tasks” baseline suffers from noise. If the “All tasks” set includes the same 1910 hard tasks plus 227 easy ones, the logic holds; if the sets are disjoint or the easy tasks dilute the signal, the causal claim needs reinforcement.
2. **Evaluator Abalation Causality (Section 4.4, Table 5):** The table compares “Decision Model” (Gemini vs. Qwen) and reports “Pass@3”. The logical leap here is whether the Pass@3 metric reflects the performance of the *adapted policy* trained with that specific evaluator, or the performance of the *base policy* re-evaluated by the new model. The text states “Replacing the final-decision model... improves Pass@3,” which suggests the policy was re-trained. If so, the causal claim is that a stronger critic yields a better policy. If the policy was fixed and only the evaluation changed, the claim is about measurement accuracy. The text implies re-training (“even the base... evaluator yields gains”), but the table header “Decision Model” is ambiguous. Clarifying that the policy was re-optimized with the new critic is essential for the logical consistency of the claim that “stronger critics could further improve safety and reliability.”
3. **Learning from Failure (Section 3.3):** The method selects the “best” attempt for training. If no attempt succeeds, it selects the one with the highest “reasonable step” count. The logical consistency of training a policy to succeed using data from *failed* trajectories relies entirely on the validity of the “process label”  $v_k^{(t)}$  as a proxy for the correct action. The paper assumes that a “reasonable” step in a failed trajectory is a correct step. This is a strong assumption. The review should note that the conclusion “we learn from failure” is only valid if the critic’s “reasonableness” label is perfectly aligned with the ground-truth action, which is not explicitly verified in the text.

Overall, the paper presents a coherent argument, but these specific logical links in the ablation analysis need tighter phrasing to ensure the conclusions are fully supported by the experimental



design described.

### Action items.

- **[writing]** Task Filtering Logic (Section 3.3, Table 4): The paper argues that retaining tasks with success rates (SR) in  $[0.0, 0.9]$  yields better out-of-domain (MobileWorld) performance than including all tasks ( $[0.0, 1.0]$ ). The logic implies that removing “mastered” tasks ( $SR=1$ ) improves generalization. However, the table shows the “Medium + hard” filter has 1910 samples while “All tasks” has 2137. The conclusion that the *filtering strategy* (excluding easy tasks) is the cause of the MobileWorld gain
- **[writing]** Evaluator Abalation Causality (Section 4.4, Table 5): The table compares “Decision Model” (Gemini vs. Qwen) and reports “Pass@3”. The logical leap here is whether the Pass@3 metric reflects the performance of the *adapted policy* trained with that specific evaluator, or the performance of the *base policy* re-evaluated by the new model. The text states “Replacing the final-decision model... improves Pass@3,” which suggests the policy was re-trained. If so, the causal claim is that a stronger cri
- **[writing]** Learning from Failure (Section 3.3): The method selects the “best” attempt for training. If no attempt succeeds, it selects the one with the highest “reasonable step” count. The logical consistency of training a policy to succeed using data from *failed* trajectories relies entirely on the validity of the “process label”  $v_k^{(t)}$  as a proxy for the correct action. The paper assumes that a “reasonable” step in a failed trajectory is a correct step. This is a strong assumption. The review shoul

## Overreach

*Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_overreach.md

The paper makes several claims regarding the “strongest” performance and the degree of “annotation-free” adaptation that slightly exceed the evidence provided in the results tables.

First, the Abstract and Conclusion assert that the method establishes the “strongest open-data mobile GUI agent in our evaluation.” While the adapted ForgeOwl-8B (41.0%) outperforms other open-data 8B models, the ForgeQwen3-8B variant achieves only 10.3% on the MobileWorld out-of-domain benchmark. This is substantially lower than the 17.7% reported for OpenMobile-8B (Table 2, Section 4.3). By grouping these results under a single “strongest” claim without qualification, the paper risks misleading readers about the generalizability of the generalist model. The claim should be refined to specify “strongest among 8B models adapted via annotation-free methods” or similar, acknowledging that other open-data baselines (like OpenMobile) perform better on out-of-domain tasks without this specific adaptation pipeline.

Second, the Abstract states that the adapted Qwen3-VL-8B (67.2%) is “close to” the closed-data GUI-Owl-1.5-8B base (69.0%). This framing obscures the fact that the *adapted* GUI-Owl-

1.5-8B (ForgeOwl-8B) achieves 77.6%, a significant 8.6 percentage point improvement over its base. The narrative suggests the adaptation brings the generalist up to the level of the *base* specialized model, but the specialized model itself benefits greatly from the same adaptation. The text should clarify that the adaptation pipeline improves both models, and the “close to” comparison is specifically between the adapted generalist and the *unadapted* specialized base, not the adapted specialized model.

Finally, the repeated emphasis on “annotation-free” adaptation (Abstract, Introduction) implies a complete lack of human supervision. However, the task generation relies on MobileGym exploring specific apps within the AndroidWorld and MobileWorld benchmarks, which are human-curated environments with predefined task structures. While the *rollout data* is not manually labeled, the *task definitions* and *evaluation metrics* are human-defined. The paper should temper claims of “fully annotation-free” to “annotation-free training data generation” to accurately reflect that the system operates within a human-structured evaluation framework rather than a truly open, unstructured environment.

#### Action items.

- **[writing]** The claim of establishing the ‘strongest open-data mobile GUI agent’ is overreaching given ForgeQwen3-8B’s 10.3% MobileWorld score, which is lower than OpenMobile-8B’s 17.7%. Qualify this claim to specify ‘strongest among 8B models’ or ‘strongest using annotation-free adaptation’ to avoid misleading readers.
- **[writing]** The abstract states the adapted generalist (67.2%) is ‘close to’ the closed-data base (69.0%), but omits that the adapted specialized model reaches 77.6%. This framing conflates the adapted generalist’s performance with the specialized model’s potential, overstating the generalist’s relative capability. Clarify the comparison targets.
- **[writing]** The paper claims ‘annotation-free’ adaptation, yet relies on human-curated benchmarks (AndroidWorld/MobileWorld) for task generation and evaluation. The system is not fully autonomous in an open world. Clarify that ‘annotation-free’ applies to training data generation, not the entire evaluation or task definition pipeline.

## Safety Ethics

*Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_safety\_ethics.md

The manuscript presents MobileForge, an annotation-free adaptation system for mobile GUI agents. From a safety and ethics perspective, the primary concerns revolve around the autonomy of the agent during the exploration and training phases, and the provenance of the data used to train the policy.

**Data Privacy and Consent:** In Section 3.2 (“MobileGym: Building an Adaptation Substrate”), the authors describe “Target-App Exploration” using depth-first traversal to collect

transition records. While the paper mentions using 20 apps (Table in Appendix e002), it does not explicitly state whether these apps contain real user data or if the exploration was conducted in a strictly controlled, synthetic environment. If any real user data (screenshots, logs, or interaction traces) was captured, the manuscript must include a statement regarding IRB approval or explicit user consent. If the data is synthetic or from public datasets, this distinction should be clearly articulated to avoid ambiguity regarding privacy violations.

**Safety of Autonomous Actions:** The core methodology involves the agent performing rollouts and receiving feedback to optimize its policy (Section 3.3, HiFPO). The “corrective hints” and “process labels” are generated automatically. There is a risk that without explicit safety constraints, the agent could learn to perform harmful actions (e.g., deleting critical files, initiating financial transactions, or bypassing security prompts) if the reward signal (task completion) is achieved through such means. The “Limitations” section (Section 6) briefly mentions dependence on the evaluator’s quality but does not detail specific safety guardrails or “kill switches” implemented during the training loop to prevent the agent from executing dangerous actions. A brief description of the safety filtering mechanisms or the scope of allowed actions would be necessary.

**Evaluation Environment:** The benchmarks (AndroidWorld, MobileWorld) are cited as the evaluation ground. It is crucial to confirm that these environments are isolated sandboxes. The review should verify that the agent’s actions during evaluation cannot propagate to real-world systems, affect other users, or cause unintended side effects outside the controlled testbed. The current text implies this but lacks an explicit safety statement regarding the isolation of the experimental setup.

**Dual-Use Considerations:** While the paper focuses on improving agent performance, the capability to autonomously navigate and manipulate mobile interfaces without human supervision has dual-use potential (e.g., automated fraud, spam generation, or privacy invasion). The authors should briefly acknowledge these potential misuse scenarios and discuss any mitigations or ethical guidelines they followed in the development of the system.

The paper is technically sound in its approach, but these safety and ethical clarifications are required before the work can be considered fully responsible for publication.

#### Action items.

- **[writing]** The manuscript presents MobileForge, an annotation-free adaptation system for mobile GUI agents. From a safety and ethics perspective, the primary concerns revolve around the autonomy of the agent during the exploration and training phases, and the provenance of the data used to train the policy. Data Privacy and Consent: In Section 3.2 (“MobileGym: Building an Adaptation Substrate”), the authors describe “Target-App Exploration” using depth-first traversal to collect transition records. While t

## Scientific Evidence

*Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_scientific\_evidence.md

The paper presents a compelling framework for annotation-free adaptation, but the scientific evidence supporting the magnitude and robustness of the reported gains requires strengthening. While the absolute performance numbers are impressive, the lack of statistical rigor in the experimental section is a significant gap.

First, the primary results in Table 1 (AndroidWorld) and Table 2 (MobileWorld) are presented as single-point estimates (e.g., 67.2% Pass@3). In reinforcement learning and policy optimization, performance can vary significantly based on random seeds, initialization, and the specific subset of tasks sampled. Without reporting standard deviations over multiple independent runs (e.g., 3-5 seeds) or confidence intervals, it is impossible to determine if the observed improvements (e.g., the +9.0% relative gain on MobileWorld) are statistically significant or within the noise floor of the training process. The claim of “strongest open-data mobile GUI agent” relies on these point estimates being robust, which is currently unproven.

Second, the ablation studies suffer from similar issues. In Table 5 (Evaluator model ablation), the difference between the Gemini 2.5 Pro evaluator (71% Pass@3) and the Qwen3-VL-8B evaluator (70% Pass@3) is marginal (1 percentage point) and based on a small sample size (200 tasks). Without error bars or a statistical test, this difference is indistinguishable from random fluctuation. Similarly, Table 4 shows that changing the task filtering strategy results in identical AndroidWorld scores (48.3%) but divergent MobileWorld scores. This suggests the AndroidWorld metric may lack the sensitivity to detect the nuances of the training data changes, or the results are unstable.

Finally, the “Curriculum grounding” ablation (Table 6) compares functional coverage percentages but does not link these coverage metrics directly to performance gains with statistical backing. While the shift in distribution is clear, the causal link between specific functional coverage increases and the final success rates needs more rigorous quantification beyond the provided tables.

To meet the standards of scientific evidence, the authors should re-run key experiments with multiple seeds to report variance, or at minimum, provide a power analysis or bootstrap confidence intervals for the reported metrics. The current evidence supports the *direction* of the improvement but not the *statistical certainty* of the claims.

#### Action items.

- **[science]** Report statistical significance (e.g., p-values, confidence intervals, or standard deviations over multiple seeds) for the key performance gains in Table 1 (AndroidWorld) and Table 2 (MobileWorld). Single-run point estimates (e.g., 67.2% vs 69.0%) are insufficient to claim robust improvement given the variance typical in RLHF/GRPO training.
- **[science]** Clarify the sample size and selection bias in the ‘Evaluator model’ ablation (Table 5). The claim that Gemini 2.5 Pro yields higher Pass@3 (71%) than Qwen3-VL-8B (70%) is based on a single run of 200 tasks. Provide variance estimates or a larger sample size to confirm this difference is not due to stochasticity or task-specific noise.
- **[science]** The ‘Task filtering’ ablation (Table 4) shows identical AndroidWorld performance

(48.3%) for two different filtering strategies but different MobileWorld results. Explain the statistical stability of this metric; a 0.0% difference in the primary metric despite a change in training data composition suggests potential overfitting or lack of sensitivity in the evaluation protocol.

## Statistical Analysis

*Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_statistical\_analysis.md

The statistical analysis in the manuscript is largely descriptive, relying on point estimates (Pass@k, success rates) without measures of uncertainty or formal hypothesis testing. While the performance gains are numerically substantial (e.g., +25.0 pp in Table 3), the lack of confidence intervals (CIs) or standard deviations (SDs) makes it difficult to assess the reliability of these improvements, particularly given the relatively small test sets (116 tasks for AndroidWorld, 117 for MobileWorld).

In **Table 1 (AndroidWorld results)** and **Table 2 (MobileWorld results)**, the authors report single-point percentages (e.g., 77.6% vs 69.0%). For binary outcomes (success/failure) on a fixed set of tasks, the standard error can be approximated as  $\sqrt{p(1-p)/n}$ . For  $n = 116$  and  $p \approx 0.7$ , the 95% CI width is roughly  $\pm 8.5\%$ . The authors should explicitly report these intervals or the number of seeds used if results are averaged over multiple runs. Without this, claims of “strongest open-data performance” are statistically weak.

Furthermore, **Section 4.3 (Ablations)** presents comparisons (e.g., Hint vs. No Hint) but does not specify the statistical significance of the observed gains. A paired test (e.g., McNemar’s test) is appropriate here since the same tasks are evaluated under different conditions. The absence of p-values or effect sizes leaves the reader unable to distinguish between genuine methodological improvements and random variance.

Regarding the **GRPO optimization (Section 3.3)**, the reward normalization (Eqs. 10-12) and KL regularization (Table 4) are described, but the statistical properties of the advantage estimator are not discussed. Specifically, the variance of the advantage estimates and the stability of the KL penalty across training steps (Figure 5) should be quantified. The training curves in **Figure 5** show reward trends but lack error bands (e.g., shaded regions for SD), which are standard for RL training analysis to demonstrate convergence stability.

Finally, the **task filtering ablation (Table 5)** compares SR ranges but does not account for the variance in sample sizes (1910 vs 2137 samples). A statistical justification for why the [0.0, 0.9] range yields better generalization is needed beyond the observed point estimates.

To improve the statistical rigor, the authors should: (1) add CIs or SDs to all performance tables; (2) report p-values for key ablation comparisons; and (3) include error bands in training curves.

**Action items.**

- **[science]** Report confidence intervals or standard deviations for Pass@k and success rates in Tables 1 and 2. Single-point estimates from 116/117 tasks lack statistical context for significance claims.
- **[science]** Clarify the statistical test used to compare baselines (e.g., McNemar’s test for paired binary outcomes). The paper claims ‘strongest’ performance without p-values or effect sizes.
- **[science]** Define the variance estimation method for the GRPO advantage estimator (Eq. 12). Specify if group normalization uses population or sample variance, and how KL penalty variance is controlled.

## Writing Quality

*Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_writing\_quality.md

The manuscript demonstrates a high level of technical sophistication, but the writing quality requires minor revisions to ensure maximum clarity and consistency. The prose is generally dense and academic, which is appropriate for the venue, but several areas suffer from inconsistent formatting and overly complex sentence structures that impede quick comprehension.

First, there is a notable inconsistency in how quantitative results are presented in the main tables. Table 1 (AndroidWorld) presents success rates as a combination of raw counts and percentages (e.g., “47/116 (40.5%)”), whereas Table 2 (MobileWorld) and the ablation tables often use only percentages. This inconsistency forces the reader to mentally adjust their parsing of the data across different sections. Standardizing this format—either always showing counts and percentages or just percentages with a clear header—would significantly improve readability.

Second, the captions for several figures, particularly Figure 3 (the HIFPO overview), are overly dense. The current caption attempts to explain the entire methodology in a single block of text, listing four distinct mechanisms (hint-guided rollout, task removal, step selection, and GRPO). This makes it difficult for a reader to quickly grasp the figure’s primary contribution. Breaking this into distinct sentences or a structured list would enhance the flow and allow the visual to stand on its own more effectively.

Third, there are minor issues with hyphenation and compound modifiers. The term “target-app” appears with a hyphen in Section 3.2 but varies in other sections. While “annotation-free” is consistently hyphenated, “target-app” should be standardized to match the manuscript’s style guide (likely using an en-dash or no hyphen depending on the specific style, but consistency is key).

Finally, the transition between the problem setup in Section 3.1 and the detailed method in 3.2 is slightly abrupt. The introduction of the decision state  $s_k^{(t)}$  and the rollout  $\tau_k$  is mathematically precise but could benefit from a brief, plain-English sentence summarizing the intuition before

diving into the equations. This would help readers who are less familiar with the specific notation to follow the narrative flow more easily.

Overall, the paper is well-written, but addressing these specific points regarding consistency and caption density will elevate the readability to match the quality of the research.

#### **Action items.**

- **[writing]** In Section 3.2, the phrase ‘Target-App Exploration’ uses a hyphen where an en-dash or no hyphen is standard for compound modifiers in this context. Ensure consistent hyphenation for ‘target-app’ throughout the manuscript (e.g., Abstract, Intro) to match the style of ‘annotation-free’.
- **[writing]** Table 1 (AndroidWorld results) and Table 2 (MobileWorld results) use inconsistent formatting for the ‘Pass@k’ and ‘SR’ columns. Table 1 mixes raw counts and percentages (e.g., ‘47/116 (40.5%)’), while Table 2 uses only percentages. Standardize the presentation of success metrics across all tables for better readability.
- **[writing]** The caption for Figure 3 (hifpo) is extremely dense and contains multiple distinct concepts (hint-guided rollout, task removal, step selection, GRPO). Consider splitting the caption into two sentences or using a bulleted list to improve clarity and flow.
- **[writing]** In Section 4.3 (Ablations), the text references ‘Table~\ref{tab:hint\_rollout\_ablation\_main}’ but the table caption itself is ‘Hint ablation (200 tasks, Qwen3-VL-8B)’. Ensure the table caption explicitly mentions the metric being ablated (e.g., ‘Corrective hints’) to align with the section header and improve skimmability.